AI for Research

Workshop starts at 09:32

Materials for today:

https://github.com/nuitrcs/coding_with_ai_assistance

Coding with AI Assistance

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About Us

This workshop is brought to you by:

Northwestern IT
Research Computing and Data Services

Need help?

- AI, Machine Learning, Data Science
- Statistics
- Visualization
- Collecting web data (scraping, APIs), text analysis, extracting information from text
- Cleaning, transforming, reformatting, and wrangling data
- Automating repetitive research tasks
- Research reproducibility and replicability
- Programming, computing, data management, etc.
- R, Python, SQL, MATLAB, Stata, SPSS, SAS, etc.

Request a **FREE** consultation at bit.ly/rcdsconsult.

Collaborative project support

If you're a faculty (or postdoc, research staff, or graduate student with faculty sponsorship), we can provide longer-term support in your project.

We can help from exploration (is this idea possible?) to concrete outcomes (e.g., data collection, cleaning, analysis or visualization for a publication).

Find more information at bit.ly/collaborative_project_support

Bring Your Own Data (BYOD)

Do you have research data waiting to be collected, cleaned, analyzed or visualized? Come to BYOD.

BYOD is a weekly working group that provides structure, guidance, and accountability to help you make progress on your research project in a supportive and stress-free environment.

Find more information at bit.ly/byod_groups

AI has an environmental impact 🌍

- Carbon footprint due to energy consumption both during model training and usage.
- Fresh water evaporated to cool servers.
- E-waste containing hazardous substances generated with the production of hardware.

Overview of the Day

What we’re doing here

- Human-in-the-loop coding assistance
- Using web-browser based chatbots, not agents
  - What are you using?
- Tool-agnostic – but you need access to at least one chatbot (can be free) and know how to use R or Python

Agenda

- Context from consults/real-world stories
- Key best practices
- Group exercise
    - Plan the project
    - Explore and clean the data
    - Work with existing code
    - Analyze and visualize
    - Group readouts
- Final debrief and takeaways

What's next?

- Tomorrow: Get Started Safely with AI Coding Agents
- Tomorrow: Linking VS Code to LLMs
- Thursday: AI for Data Visualization
- Thursday: Workflow Strategies for AI Agents

- November 19: Using LLM APIs through Azure
- November 19: Using LLMs on Quest

- Talks about practical applications of AI in research in the fall!

What We Are Seeing

SLOP NIGHTMARE

A Scientist Was Presenting His Vibe-Coded New Work When One of His Colleagues Pointed Out Something Extremely Embarrassing

"It was subtle, but it was very wrong, and everything downstream of it was also wrong, and I had shared the whole thing in a room full of people who trusted me."

https://futurism.com/artificial-intelligence/scientist-horrified-vibe-coded-work-conference-wrong

The dark side

- Figure created “without code”
- Moving web-app from local machine to GitHub Actions
- Pipeline with low accuracy

Do you have any other stories?

The bright side

- Planning and scaffolding
- Prototyping
- UIs to make review easier
- Information extraction
- Better research software practices

Do you have any other stories?

Key Best Practices

Main messages

- Ask for code, not conclusions/analysis
- Break it down yourself
- Human in the loop – you drive
- Research and data practices still apply

Ask for code, not conclusions/analysis

- Have the AI write code you run yourself — don't ask it to analyze your data or hand you conclusions directly
- The code an AI writes is trained for software engineering rather than research data analysis
  - Tends to be verbose and overly complicated
  - Structured differently

Break it down yourself

- Break the problem into small chunks before you prompt
- Ask the AI to write a plan (and/or pseudocode) and check in with you before it generates code
- Ask it to state its assumptions or ask clarifying questions first — don't let it guess
- Give it good context — a vague prompt gets a vague answer

Human in the loop – you drive

- Review and understand the code before you run it
- Cross-check unfamiliar library usage against the documentation
- Iterate — one prompt rarely gets it right; use follow-ups or a fresh conversation

Research and data practices still apply

- Know your data before and while you clean/analyze it
- Ask the AI to help think through edge cases and failure modes, then verify them yourself
- Keep code complexity matched to the project's scale and goals
- Verify that the input and output is as expected in each step of the code
- Document your work and give credit — comments/docstrings/README, which AI tool you used

Main messages

- Ask for code, not conclusions/analysis
- Break it down yourself
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- Research and data practices still apply

We have more best practices here: https://github.com/nuitrcs/coding_with_ai_assistance/blob/main/best_practices.md

Exercise Kickoff

High-level overview of the exercise

- You’ll be working in groups of 3-4 people
- Each group will decide a track – selecting a research question and dataset from the social sciences, life sciences, or engineering
- For most of the rest of the day you’ll be working on planning the project, exploring and cleaning the data, working with existing code, and analyzing and visualizing the data – with AI help
- By the end of the day each group will give a brief readout (3 min)
- Discuss and compare prompts, tools, and outputs between each other
- *We’ll be circulating around to help!*

Context for the exercises

- The datasets for the exercises come from the replication packages of three recently published articles
- The research questions that we are providing for you today are not the full questions answered in the articles
- We are also not asking you to work with the full data for the articles
- If you finish early, feel free to go to the full replication package, use AI to understand it, and extend your work

Let’s get started!

- Choose a track – social sciences, life sciences, or engineering
- Form a group of 3-4 people
- Open the README.md in your track folder (which has the research question and data)
- Start working!
- *We are here to help you!*

All the materials are here: https://github.com/nuitrcs/coding_with_ai_assistance/tree/main/exercise

Debrief and Takeaways

Main messages

- Ask for code, not conclusions/analysis
- Break it down yourself
- Human in the loop – you drive
- Research and data practices still apply

We have more best practices here:
https://github.com/nuitrcs/coding_with_ai_assistance/blob/main/best_practices.md

What best practices did you try today?

Did you try any best practices that we didn’t cover?

What are three best practices that you learned today that you're planning to try in your own work?